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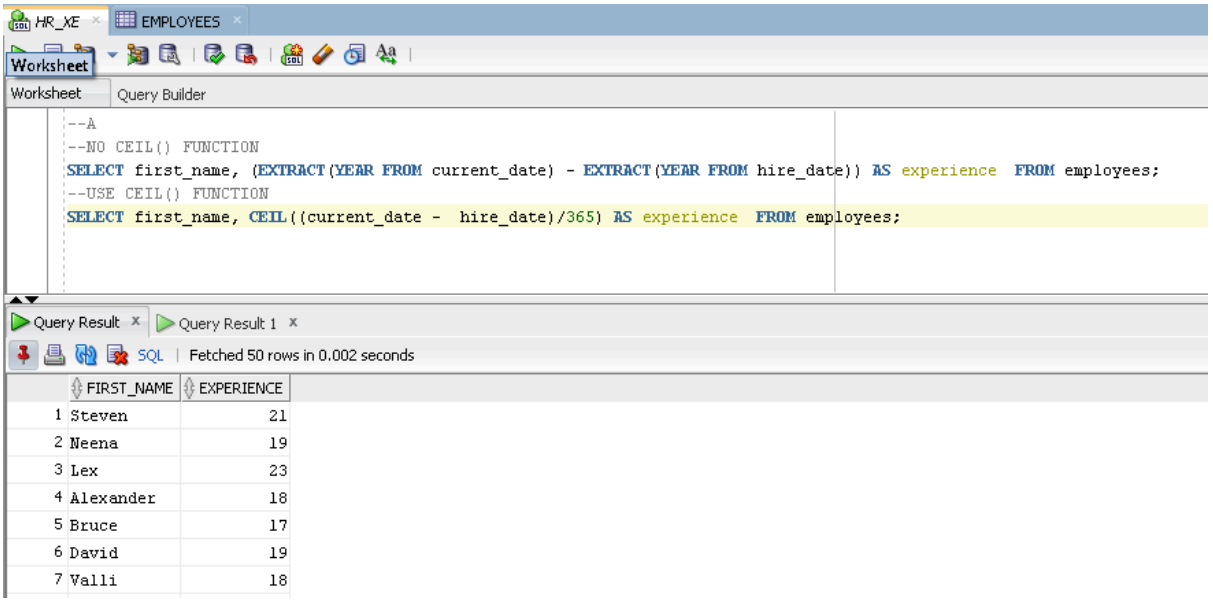
Lab 2. Accessing Table Data, DML Statements, and Transactions

Write SQL queries to:

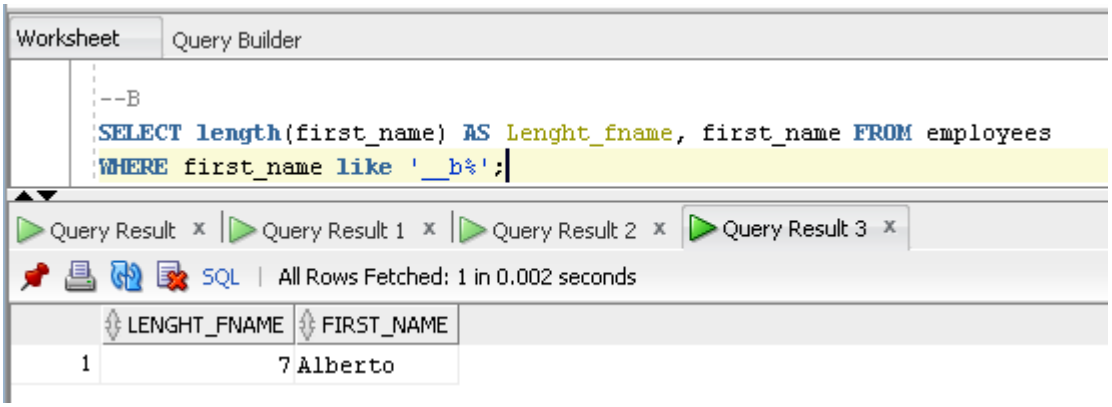
a. Display first name and experience of the employees.

```
--NO CEIL() FUNCTION
SELECT first_name, (EXTRACT(YEAR FROM current_date) - EXTRACT(YEAR FROM hire_date)) AS experience  FROM employees;
--USE CEIL() FUNCTION
SELECT first_name, CEIL((current_date -  hire_date)/365) AS experience  FROM employees;
```



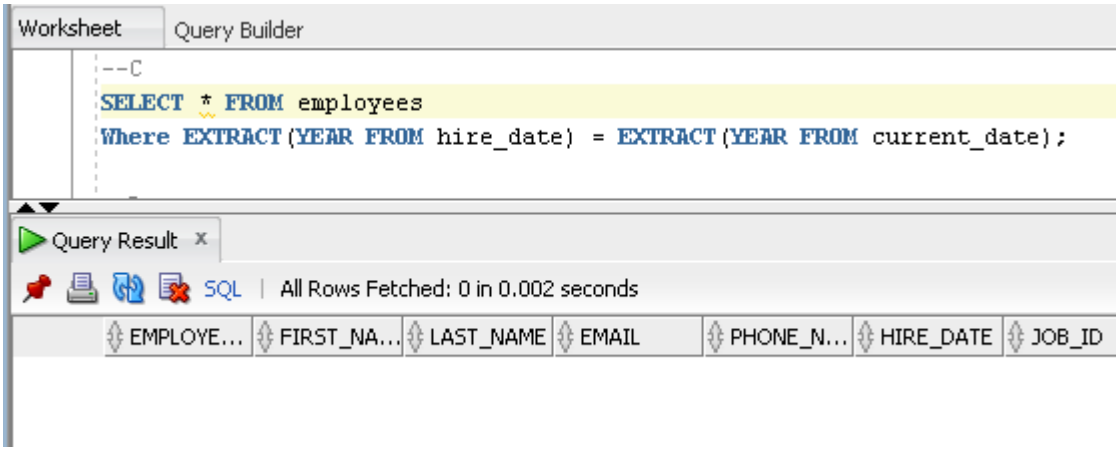
b. Display the length of first name for employees where last name contain character ‘b’ after 3rd position.

```
SELECT length(first_name) AS Lenght_fname, first_name FROM employees
WHERE first_name like '__b%';
```



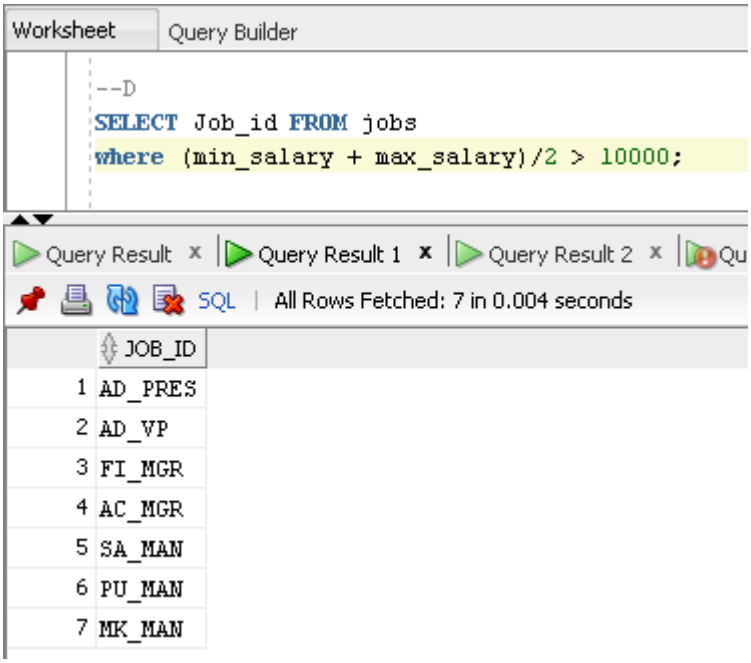
c. Display employees who joined in the current year.

SELECT * FROM employees
Where EXTRACT(YEAR FROM hire_date) = EXTRACT(YEAR FROM current_date);



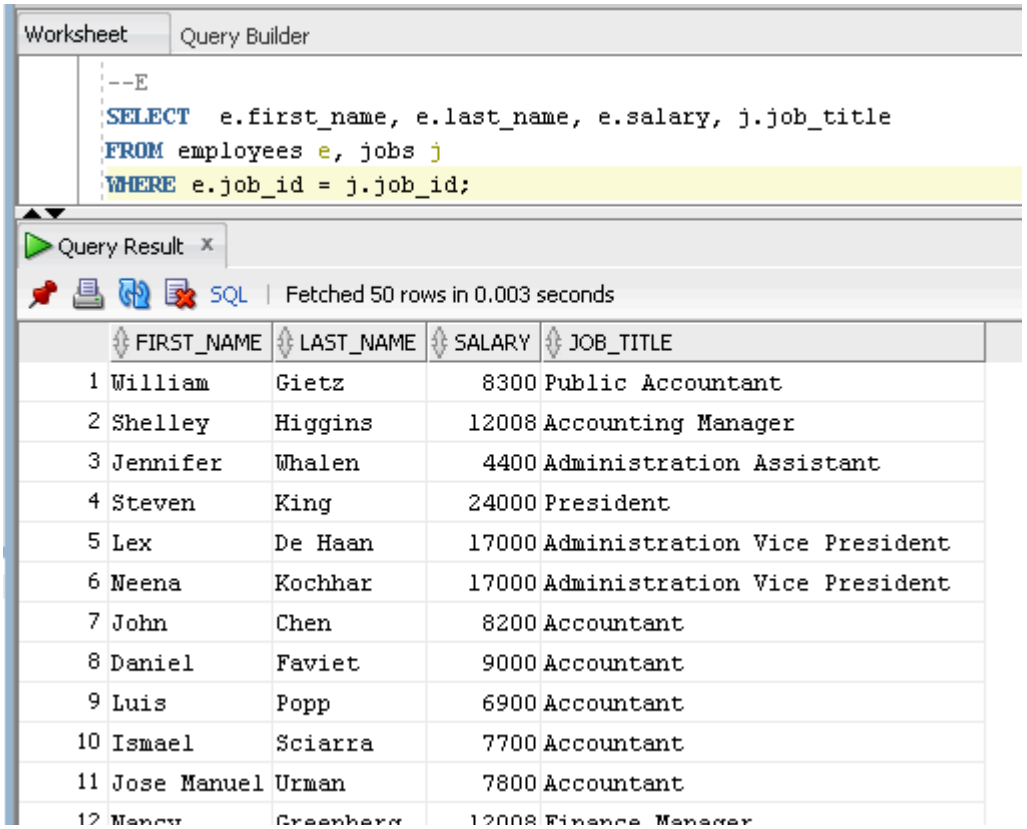
d. Display job ID for jobs with average salary more than 10,000.

SELECT Job_id FROM jobs
where (min_salary + max_salary)/2 > 10000;



e. Display the first name, last name, salary and job grade for all employees.

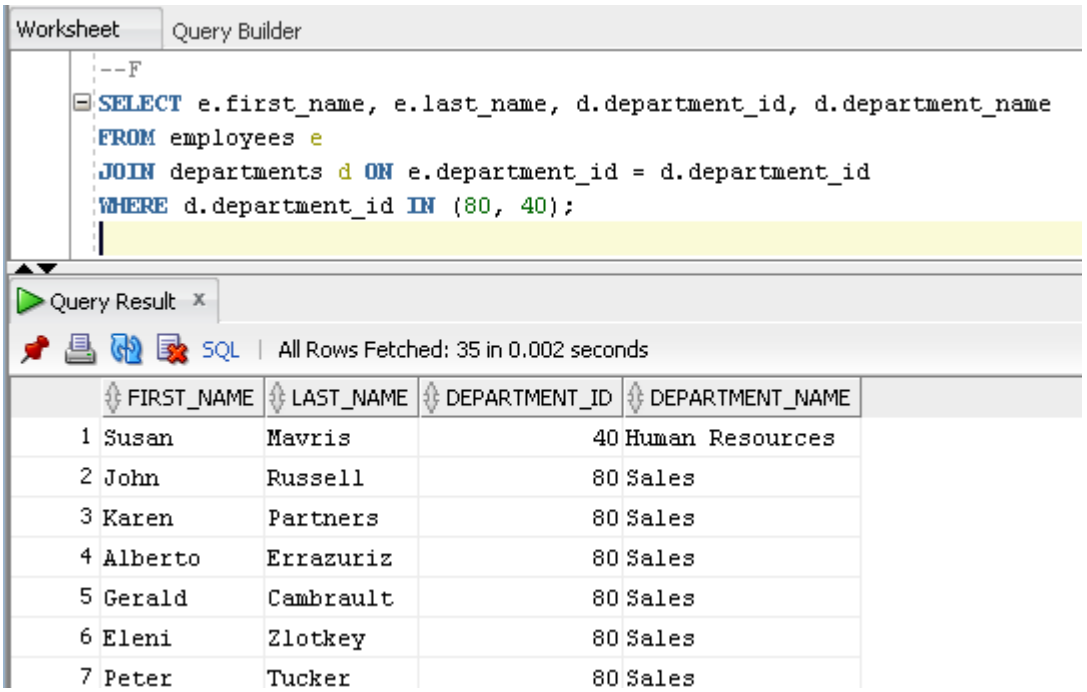
SELECT e.first_name, e.last_name, e.salary, j.job_title
FROM employees e, jobs j
WHERE e.job_id = j.job_id;



	FIRST_NAME	LAST_NAME	SALARY	JOB_TITLE
1	William	Gietz	8300	Public Accountant
2	Shelley	Higgins	12008	Accounting Manager
3	Jennifer	Whalen	4400	Administration Assistant
4	Steven	King	24000	President
5	Lex	De Haan	17000	Administration Vice President
6	Neena	Kochhar	17000	Administration Vice President
7	John	Chen	8200	Accountant
8	Daniel	Faviet	9000	Accountant
9	Luis	Popp	6900	Accountant
10	Ismael	Sciarra	7700	Accountant
11	Jose Manuel	Urman	7800	Accountant
12	Nancy	Greenberg	12008	Finance Manager

f. Display the first name, last name, department number and department name for all employees for departments 80 or 40.

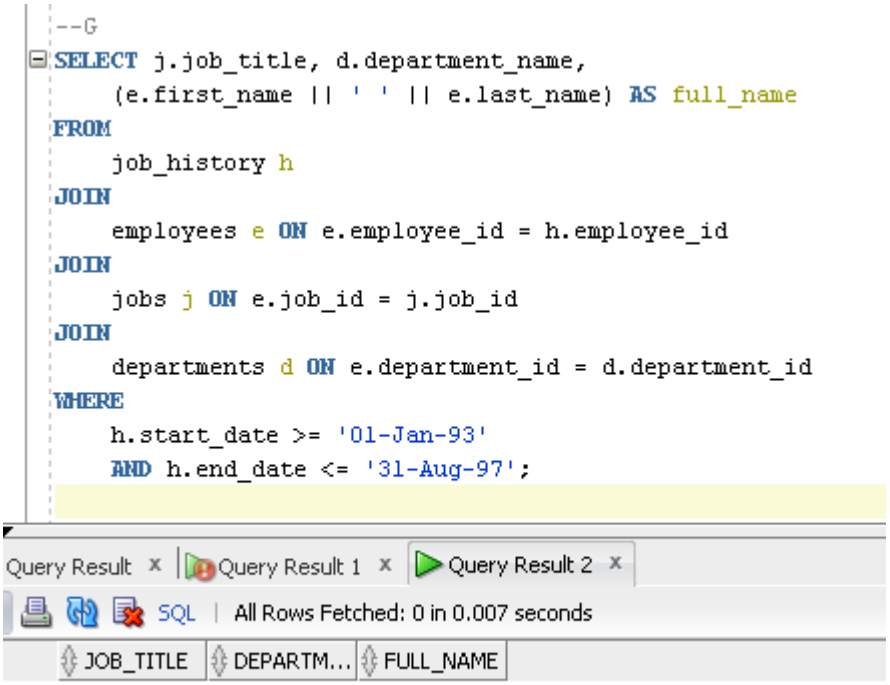
SELECT e.first_name, e.last_name, d.department_id, d.department_name
FROM employees e
JOIN departments d ON e.department_id = d.department_id
WHERE d.department_id IN (80, 40);



	FIRST_NAME	LAST_NAME	DEPARTMENT_ID	DEPARTMENT_NAME
1	Susan	Mavris	40	Human Resources
2	John	Russell	80	Sales
3	Karen	Partners	80	Sales
4	Alberto	Errazuriz	80	Sales
5	Gerald	Cambrault	80	Sales
6	Eleni	Zlotkey	80	Sales
7	Peter	Tucker	80	Sales

g. Display the job title, department name, full name (first and last name) of employee and starting date for all the jobs which started on or after 1st January, 1993 and ending with on or before 31 August, 1997.

```
SELECT j.job_title, d.department_name,
   (e.first_name || ' ' || e.last_name) AS full_name
FROM
   job_history h
JOIN
   employees e ON e.employee_id = h.employee_id
JOIN
   jobs j ON e.job_id = j.job_id
JOIN
   departments d ON e.department_id = d.department_id
WHERE
   h.start_date >= '01-Jan-93'
   AND h.end_date <= '31-Aug-97';
```



h. Display the department name and number of employees in each of the department.

```
SELECT d.department_name, count(*) AS num_employees
FROM employees e
JOIN departments d ON e.department_id = d.department_id
GROUP BY d.department_id, d.department_name;
```

```
--H
SELECT d.department_name, count(*) AS num_employees
FROM employees e
JOIN departments d ON e.department_id = d.department_id
GROUP BY d.department_id, d.department_name;
```

Query Result | SQL | All Rows Fetched: 11 in 0.004 seconds

	DEPARTMENT_NAME	NUM_EMPLOYEES
1	Finance	6
2	Shipping	45
3	Public Relations	1
4	Purchasing	6
5	Executive	3
6	Administration	1
7	Accounting	2
8	Human Resources	1
9	Marketing	2
10	IT	5
11	Sales	34

i. Display years in which more than 10 employees joined.

SELECT EXTRACT(YEAR FROM hire_date)-- count(*)
FROM employees
GROUP BY EXTRACT(YEAR FROM hire_date)
HAVING count(*)>10;

Worksheet | Query Builder

```
--I
SELECT  EXTRACT(YEAR FROM hire_date)-- count(*)
FROM employees
GROUP BY  EXTRACT(YEAR FROM hire_date)
HAVING count(*)>10;
```

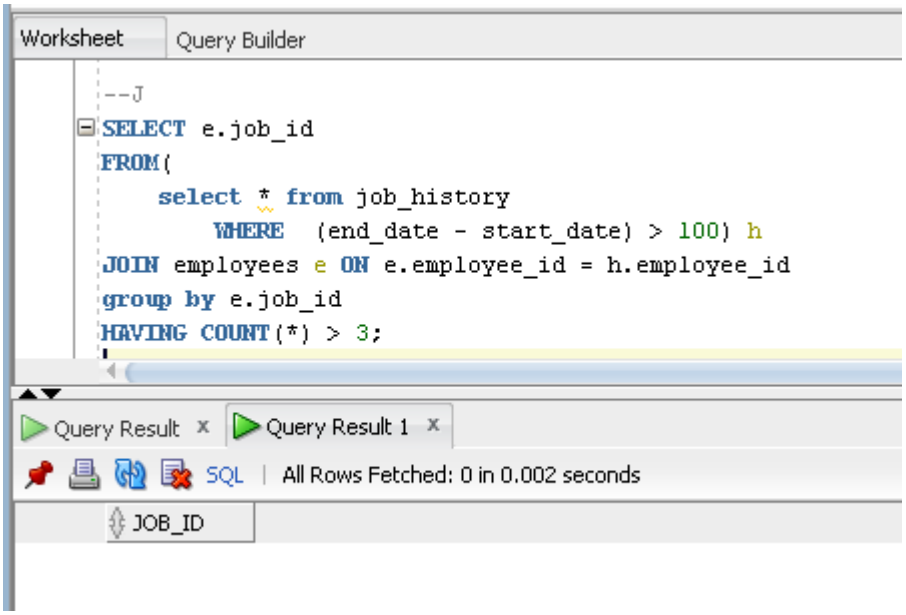
Query Result | SQL | All Rows Fetched: 4 in 0.002 seconds

	EXTRACT(YEARFROMHIRE_DATE)	COUNT(*)
1	2005	29
2	2006	24
3	2007	19
4	2008	11

j. Display job ID of jobs that were done by more than 3 employees for more than 100 days.

SELECT e.job_id
FROM(

```
    select * from job_history
       WHERE  (end_date - start_date) > 100) h
JOIN employees e ON e.employee_id = h.employee_id
group by e.job_id
HAVING COUNT(*) > 3;
```



k. Change job ID of the employee 110 to IT_PROG if the employee belongs to department 10 and the existing job ID does not start with IT.

```
UPDATE employees
SET job_id = 'IT_PROG'
WHERE employee_id = 110
  AND department_id = 10
  AND job_id NOT LIKE 'IT%';
```